

Bayesian Probability Analysis

IMDb Movie Reviews: Sentiment via Bayes' Theorem

1. Keyword Selection

We selected three keywords that we supposed signal positive sentiment and three that signal negative sentiment, and one control word that appears in both classes. Every count in this report is directly from the dataset meaning that no values were assumed.

Positive keywords

Keyword	Justification
masterpiece	Reserved for films the reviewer rates as exceptional.
wonderful	Broad, strong term of praise for the overall experience.
brilliant	High praise for writing, acting, or direction.

Negative keywords

Keyword	Justification
boring	Direct signal of disengagement and lack of interest.
disappointing	The film failed to meet the reviewer's expectations.
waste of time	Strong dismissal implying no value in watching.

Control keyword

Keyword	Justification
funny	Appears in both praise and criticism so we used it as a sanity check.

2. Choice of Conditional Probability

Our group calculates $P(\text{Positive} \mid \text{keyword})$ — "given that this word appears in a review, how likely is the review positive?" Per the assignment, we compute this one direction only (not $P(\text{Negative} \mid \text{keyword})$). It is the most directly interpretable framing for predicting sentiment from a keyword.

3. Dataset Statistics

Statistic	Value
Total reviews	50,000
Positive reviews	25,000 (50%)

Negative reviews	25,000 (50%)
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Since the classes are perfectly balanced, so the prior is:

$$P(\text{Positive}) = 25,000 / 50,000 = 0.50$$

4. Keyword Frequency (reviews containing the keyword)

Keyword	In positive	In negative	Total
masterpiece	878	329	1207
wonderful	2258	522	2780
brilliant	1587	501	2088
boring	591	2460	3051
disappointing	171	615	786
waste of time	30	661	691
funny	3089	3286	6375

5. Bayesian Calculations

Formula:

$$P(\text{Positive} \mid \text{keyword}) = [P(\text{keyword} \mid \text{Positive}) \times P(\text{Positive})] / P(\text{keyword})$$

For each keyword: the likelihood is $P(\text{keyword} \mid \text{Positive}) = (\text{count in positive}) / 25,000$; the marginal is $P(\text{keyword}) = (\text{total count}) / 50,000$; and the posterior follows from Bayes' Theorem above.

Positive keyword — "masterpiece"

$$P(\text{keyword} \mid \text{Positive}) = 878 / 25,000 = 0.03512$$

$$P(\text{keyword}) = 1207 / 50,000 = 0.02414$$

$$P(\text{Positive} \mid \text{"masterpiece"}) = (0.03512 \times 0.50) / 0.02414 = \mathbf{0.727}$$

Interpretation: a review containing "masterpiece" is **72.7%** likely to be positive.

Positive keyword — "wonderful"

$$P(\text{keyword} \mid \text{Positive}) = 2258 / 25,000 = 0.09032$$

$$P(\text{keyword}) = 2780 / 50,000 = 0.05560$$

$$P(\text{Positive} \mid \text{"wonderful"}) = (0.09032 \times 0.50) / 0.05560 = \mathbf{0.812}$$

Interpretation: a review containing "wonderful" is **81.2%** likely to be positive.

Positive keyword — "brilliant"

$$P(\text{keyword} \mid \text{Positive}) = 1587 / 25,000 = 0.06348$$

$$P(\text{keyword}) = 2088 / 50,000 = 0.04176$$

$$P(\text{Positive} \mid \text{"brilliant"}) = (0.06348 \times 0.50) / 0.04176 = \mathbf{0.760}$$

Interpretation: a review containing "brilliant" is **76.0%** likely to be positive.

Negative keyword — "boring"

$$P(\text{keyword} \mid \text{Positive}) = 591 / 25,000 = 0.02364$$

$$P(\text{keyword}) = 3051 / 50,000 = 0.06102$$

$$P(\text{Positive} \mid \text{"boring"}) = (0.02364 \times 0.50) / 0.06102 = \mathbf{0.194}$$

Interpretation: a review containing "boring" is **19.4%** likely to be positive.

Negative keyword — "disappointing"

$$P(\text{keyword} \mid \text{Positive}) = 171 / 25,000 = 0.00684$$

$$P(\text{keyword}) = 786 / 50,000 = 0.01572$$

$$P(\text{Positive} \mid \text{"disappointing"}) = (0.00684 \times 0.50) / 0.01572 =$$

0.218 Interpretation: a review containing "disappointing" is **21.8%** likely to be positive.

Negative keyword — "waste of time"

$$P(\text{keyword} \mid \text{Positive}) = 30 / 25,000 = 0.00120$$

$$P(\text{keyword}) = 691 / 50,000 = 0.01382$$

$$P(\text{Positive} \mid \text{"waste of time"}) = (0.00120 \times 0.50) / 0.01382 =$$

0.043 Interpretation: a review containing "waste of time" is **4.3%** likely to be positive.

Control keyword — "funny"

$$P(\text{keyword} \mid \text{Positive}) = 3089 / 25,000 = 0.12356$$

$$P(\text{keyword}) = 6375 / 50,000 = 0.12750$$

$$P(\text{Positive} \mid \text{"funny"}) = (0.12356 \times 0.50) / 0.12750 = \mathbf{0.485}$$

Interpretation: a review containing "funny" is **48.5%** likely to be positive.

6. Summary of Posteriors

Keyword	Category	P(Positive keyword)
masterpiece	Positive	72.7%
wonderful	Positive	81.2%
brilliant	Positive	76.0%
boring	Negative	19.4%
disappointing	Negative	21.8%
waste of time	Negative	4.3%
funny	Control	48.5%

7. Key Findings

- **Strongest positive predictor:** "wonderful" at 81.2%.
- **Strongest negative predictor:** "waste of time" at 4.3% (i.e. 95.7% likely negative).
- All three

positive keywords sit above the 50% prior; all three negative keywords sit well below it.

- The control word "funny" lands at 48.5% right at the baseline. Because it appears in both praise ("funny and clever") and criticism ("tried to be funny and failed"), it carries almost no sentiment on its own, which is exactly what a control word should show.

8. Conclusion

Bayes' Theorem turns raw keyword counts into a defensible probability of sentiment. The chosen positive and negative keywords behave as expected, and the control word confirms the method is not simply labelling every frequent word as positive. The full implementation reading all 50,000 reviews and computing every probability in pure Python is provided in `Isimbi_Selena_Bayesian_Code.ipynb`.